

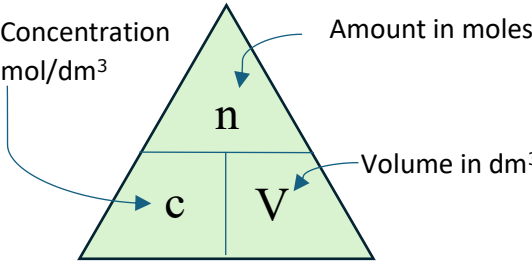
Unit 1: Structure, Bonding and Introduction to Organic Chemistry

Topic 1: Formulae, Equations and Amount of Substance

Application of ideas from this topic will be applied to all other units.

Students will be assessed on their ability to:

1.1	know the terms 'atom', 'element', 'ion', 'molecule', 'compound', 'empirical formula' and 'molecular formula'
	An atom is the smallest, electrically neutral particle of an element that can take part in a reaction An element is a substance which cannot be broken down into simpler substances by chemical means and it can contain only one type of atom An ion is an electrically charged atom or group of atoms that has gained or lost electrons to become a charged species A molecule is the smallest electrically neutral particle of an element or compound which can exist on its own and is formed by two or more atoms bonded together A compound is a pure substance which can be broken down into simpler substances by chemical means and it contains two or more different atoms (elements) chemically joined together An empirical formula shows the smallest whole number ratio of atoms of each element in a compound or molecule A molecular formula shows the actual number of atoms of each element in a compound or molecule
1.2	know that the mole (mol) is the unit for the amount of a substance and be able to perform calculations using the Avogadro constant L ($6.02 \times 10^{23} \text{ mol}^{-1}$)
	(ii) Calculate the number of nitrogen atoms in 5.60 g of nitrogen gas. 2019 Jan U1 Q-22 [Avogadro constant = $6.02 \times 10^{23} \text{ mol}^{-1}$] (2)
1.3	write balanced full and ionic equations, including state symbols, for chemical reactions
	(iii) Use your answers to (a)(i) and (a)(ii) to write the ionic equation for the reaction of iron with iron(III) chloride. Include state symbols. 2022 June U1 Q-19 You must show your working. (3)

1.4	understand the terms: - i 'relative atomic mass' based on the ^{12}C scale - ii 'relative molecular mass' and 'relative formula mass', including calculating these values from relative atomic masses <i>The term 'relative formula mass' should be used for compounds with giant structures.</i> - iii 'molar mass' as the mass per mole of a substance in g mol^{-1} - iv parts per million (ppm), including gases in the atmosphere
	Relative atomic mass (A_r) is the weighted average mass of an atom of an element, taking into account the abundance of all the isotopes of that element - It is measured as a ratio $1/12^{\text{th}}$ the mass of a carbon-12 atom Avogadro's constant (6.02×10^{23}) is the number of particles in one mole of any substance (whether it be electrons, ions, atoms or molecules) A mole is the amount of a substance that contains the same number of particles as the number of carbon atoms in exactly 12g of the ^{12}C isotope The number of carbon atoms in exactly 12g of the ^{12}C isotope is the Avogadro's constant 6.02×10^{23} Basically, a mole is just a unit of measurement like kilograms or pounds Or more similarly, just like how a dozen is 12! A mole is simply 6.02×10^{23} of anything whether it be electrons, ions, atoms, molecules or soda cans! A dozen of books is 12 books/ A mole of oxygen is 6.02×10^{23} oxygen molecules
1.5	calculate the concentration of a solution in mol dm^{-3} and g dm^{-3} <i>Titration calculations are not required at this stage.</i>
(molar) in mol/dm^3	 Concentration in ppm = $\frac{\text{mass of solute}}{\text{mass of solvent}} \times 10^6$
1.6	be able to use experimental data to calculate empirical and molecular formulae
	(b) A sample of a compound is analysed and found to contain only 3.09 g carbon, 0.26 g hydrogen and 9.15 g chlorine. The molar mass of the compound is 97.0 g mol^{-1}. Calculate the molecular formula of this compound. You must show your working. 2019 Jan U1 Q-24 (3)

1.7	be able to use chemical equations to calculate reacting masses and vice versa, using the concepts of amount of substance and molar mass	
	(e) When 6.95 g of $\text{FeSO}_4 \cdot x\text{H}_2\text{O}$ is heated, 2.00 g of iron(III) oxide, 0.80 g of sulfur dioxide and 1.00 g of sulfur trioxide are produced. The only other product is water. Deduce the overall equation for the reaction using these data. State symbols are not required. You must show your working. [A_r values: H = 1.0 O = 16.0 S = 32.1 Fe = 55.8]	2021 June U1 Q-24 (5)
1.8	be able to use chemical equations to calculate volumes of gases and vice versa, using:	
	i the concepts of amount of substance ii the molar volume of gases iii the expression $pV = nRT$ for gases and volatile liquids	
	(iii) A sample of nitrogen gas occupied 108 cm^3 at a temperature of 25°C and a pressure of $1.36 \times 10^5\text{ Pa}$. Using the ideal gas equation, calculate the number of moles of nitrogen gas in this sample. [$pV = nRT$ $R = 8.31\text{ J mol}^{-1}\text{ K}^{-1}$]	2019 Jan U1 Q-22 (4)

1.9	be able to calculate percentage yields and percentage atom economies (by mass) in laboratory and industrial processes, using chemical equations and experimental results Atom economy = $\frac{\text{molar mass of the desired product}}{\text{sum of the molar masses of all products}} \times 100$
	(ii) An experiment was carried out to produce pure, dry crystals of hydrated copper(II) sulfate, $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$. Copper(II) carbonate was mixed with 50.0 cm^3 of 1.00 mol dm^{-3} sulfuric acid until no more reacted. The mass of $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$ obtained was 10.87 g. Calculate the percentage yield for this reaction, giving your answer to an appropriate number of significant figures. [Molar mass of $\text{CuSO}_4 \cdot 5\text{H}_2\text{O} = 249.6\text{ g mol}^{-1}$] (4) (d) Germane is a compound with the formula GeH_4. It can be formed by the reaction shown. $\text{Na}_2\text{GeO}_3 + \text{NaBH}_4 + \text{H}_2\text{O} \rightarrow \text{GeH}_4 + 2\text{NaOH} + \text{NaBO}_2$ (i) Calculate the atom economy, by mass, for the formation of germane. Use A_r of Ge = 72.6 (2) 2022 Jan U1 Q-20 2022 Oct U1 Q-16

1.10	be able to determine a formula or confirm an equation by experiment, including evaluation of the data
	(d) In an experiment, 8.00 cm^3 of 0.250 mol dm^{-3} sodium hydroxide, NaOH, reacted completely with 10.0 cm^3 of 0.100 mol dm^{-3} phosphoric acid, H_3PO_4. 2019 Oct U1 Q-24 Use these data to deduce the balanced equation for this reaction. You must show your working. (3)
1.11	CORE PRACTICAL 1 Measurement of the molar volume of a gas.
1.12	be able to relate ionic and full equations, with state symbols, to observations from simple test-tube experiments, to include: - i displacement reactions - ii typical reactions of acids - iii precipitation reactions
	Further suggested practicals: - i preparation of a salt and calculating the percentage yield of product, including the preparation of a double salt, such as ammonium iron(II) sulfate from iron, ammonia and sulfuric acid - ii determine a chemical formula by experiment, such as the formula of copper(II) oxide by reduction - iii determine a chemical equation by experiment, such as the reaction between lithium and water, or the reaction between magnesium and an acid - iv carry out and interpret the results of simple test-tube reactions, outlined in 1.12